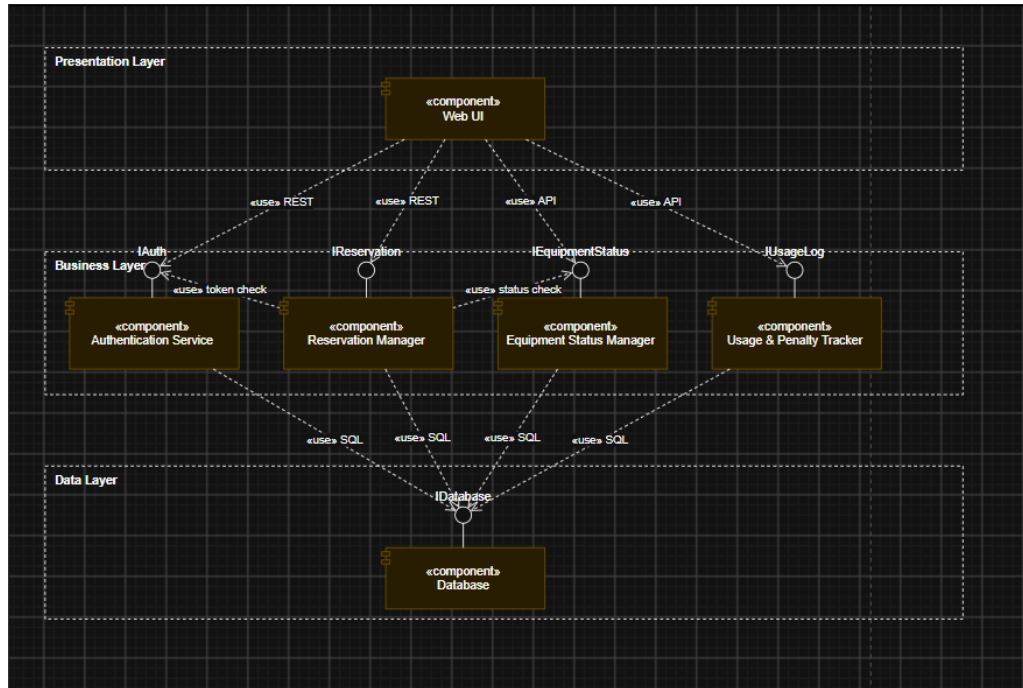


Lab 3: Component Modelling & Architectural Pattern Selection

System: Smart Lab Equipment & Slot Reservation Portal



Architecture Selection: We chose Layered Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice: The system is organised into three layers. The Presentation layer holds the Web UI. The Business layer holds the Authentication Service, Reservation Manager, Equipment Status Manager and Usage & Penalty Tracker. The Data layer holds the Database. Each layer communicates only with the layer below it through defined interfaces

Reason 1: Centralised slot locking prevents double-booking. The portal must process concurrent reservation requests without double-booking. In a layered design, all business logic reaches a single shared database, so the Reservation Manager can apply database-level locks in one place. This is simpler and more reliable than keeping data consistent across several independent services.

Reason 2: Components map directly to the requirements. Reservation and cancellation (FR-001, FR-002), equipment status and usage tracking with late-return penalties each correspond to one business component. This makes each part easy to build, test and replace independently. A campus lab with a small student user base does not need the operational complexity of microservices.

Security Advantage: Every request passes through the Authentication Service, so role-based access control is enforced in one place. Students can access only their own reservations, while Lab Technicians get equipment management functions. The Web UI never accesses the database directly, which reduces the risk of unauthorised data access.

Performance Benefit: All business components run in the same deployment and talk to one database, so there are no network hops between services. This keeps reservation processing within the 200 ms target (NFR-001), whereas a microservices design would add network latency to every request.